Electrical grid automated visual asset inspection - an example application to PV modules

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Abstract

In this paper we investigate a deep learning approach to the automated classification of defects in operating photovoltaic modules through the analysis of outdoor collected thermography images. We test several variants of a simple deep learning architecture on a publicly available dataset of labeled infrared images divided into 12 classes. The classification results yield Precision, Recall and F1-score of 0.87, 0.86 and 0.86, respectively, demonstrating the practical value of the presented automated approach.

Motivation

Globally installed solar power exceeded 1000 GW at the end of 2022 with up to 400 GW expected to be installed in 2023 alone. With the rapid expansion of solar photovoltaic (PV) plants, robust and efficient inspection methods are crucial for detecting defects in PV modules and associated systems. Thermographic imaging of PV modules, using thermal infrared (IR) cameras to detect temperature irregularities when compared to healthy ones, has emerged as a valuable inspection approach.

In the context of the rapidly expanding installation of modern PV systems, manual collection and analysis of images becomes impractical. In practice, huge amounts of images are collected by UAVs equipped with thermal cameras while automated analysis using computer vision with machine learning allows to identify patterns linked to defective modules within thermographic imagery.

Methodology

A complete solution for inspecting a PV plant starts with data acquisition, including UAV route planning, followed by video frame cropping to create thermographic images of PV modules. These images are then categorized into different defect classes, ultimately leading to the elaboration of a maintenance report. In this study we focus on the central part of this pipeline system: the computer vision core functions as a sensor that interprets physical quantities and provides valuable classification information.

Our system involves using cropped thermal images of PV modules as input data and generate predictions for each defect class, outputting the top predicted class. We tested 4 custom CNN models on the dataset in [1] containing the images presented in Figure 1.

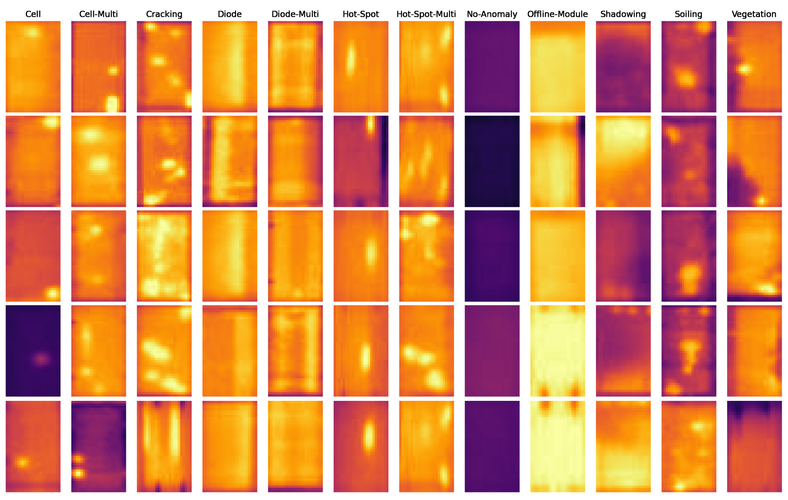


Figure 1. Example of the diversity of insulator image samples

From the original dataset, we randomly selected 150 images from each class, reserved 100 for validation and testing, and augmented the other 50 for training. The augmentation procedure involved only basic operations, such as rotation and mirroring, because other conventional transformations such as cropping and shearing, have the potential to modify crucial features. Table 1 presents the augmentation stages that ended up with a total of 6000 images across the 12 available classes (80%, 10%, 10%).

Table 1. Data selection and augmentation results

Class	Original Dataset	Random Subset	Testing & Validation	Training Split	Rotate x 2	V Flip x 2	H Flip x 2	Augmented + Test & Valid
Cell	1877	150	100	50	100	200	400	500
Cell-Multi	1288	150	100	50	100	200	400	500
Cracking	940	150	100	50	100	200	400	500
Diode	1499	150	100	50	100	200	400	500
Diode-Multi	175	150	100	50	100	200	400	500
Hot-Spot	249	150	100	50	100	200	400	500
Hot-Spot-Multi	246	150	100	50	100	200	400	500
No-Anomaly	10.000	150	100	50	100	200	400	500
Offline	827	150	100	50	100	200	400	500
Shadowing	1056	150	100	50	100	200	400	500
Soiling	204	150	100	50	100	200	400	500
Vegetation	1639	150	100	50	100	200	400	500
Total	20.000	1800	1200	600	1200	2400	4800	6000

This option resulted from the recognition that an imbalanced dataset, when used for multi-class classification, can lead to sub-optimal training outcomes. There is a risk that the model might predominantly predict the majority class for most instances, minimizing the overall loss but neglecting the minority classes, overshadowing in the learning process, and affecting the model's performance.

Architecture

Ensuring feature consistency, we maintained the original 40x24 spatial resolution and the 8-bit amplitude resolution of the images in the dataset. Our investigation involved a model architecture based on the work in [2], with structural adjustments to suit the dataset used. This architecture, referred to as M1 in Figure 3 and the benchmarking Table 2, serves as a reference for performance comparisons.

The tested models share a common structure: a block of convolutional layers followed by a block of dense layers, with a flatten layer in between. The first layer is a normalization layer, while the final dense layer utilizes *softmax* activation for classifying into the 12 predefined classes.

The convolutional layers use a kernel size of 3x3, with padding set to 'same,' and a stride of 1 to maintain output size or a stride of 2 to replace max-pooling layers. ReLU activation is applied to both Conv2D and dense layers. Figure 2 presents a comparison between the different tested architectures from model M1 to model M4.

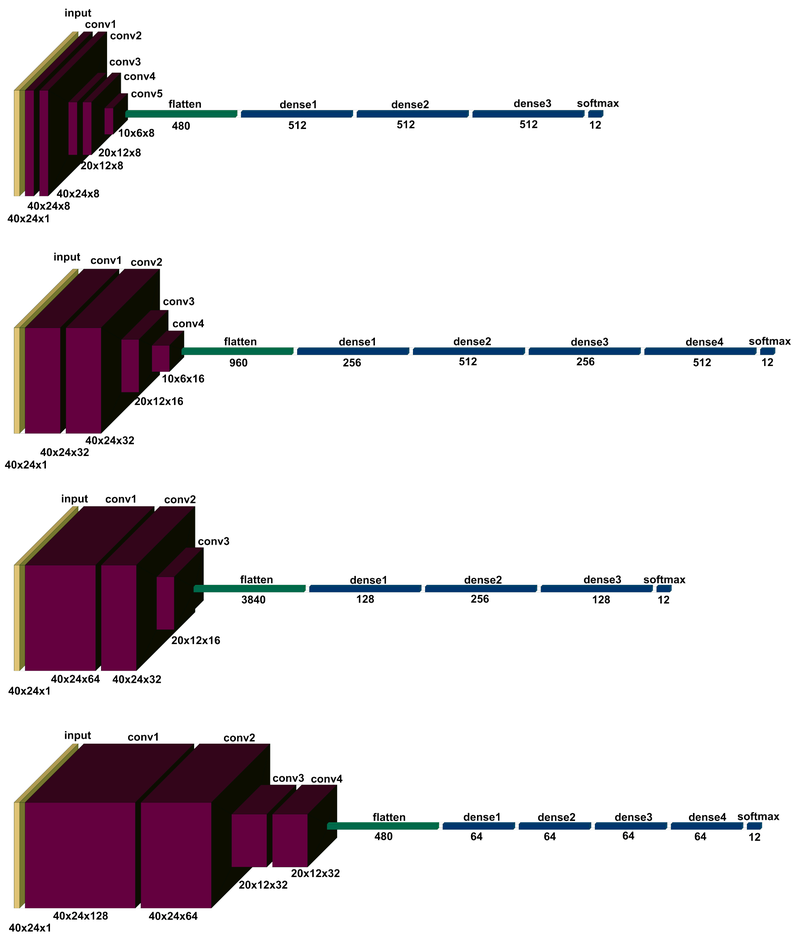


Figure 2. Tested models (M1, M2, M3, M4) architecture diagram

The leading direction was to increase the number of parameters in the convolutional block and decrease them in the dense block to hopefully achieve an optimal balance between the two.

Results

The evolution of the loss function as a result of the training procedure is presented in Figure 3. We observe that model M1 starts to overfit around the 20th epoch, while model M2 doesn't appear to overfit before the 50th epoch, but it also not converging. Model M3 begins to overfit after the 27th epoch but achieves the lowest loss value. M4 overfits around the 34th epoch, however does not outperform M3.

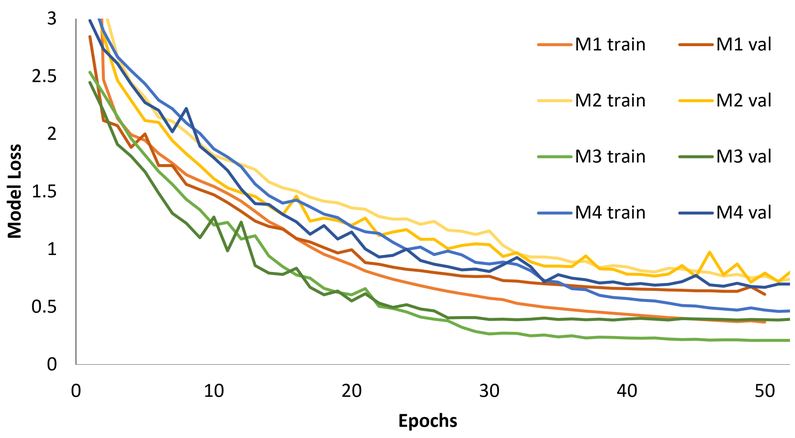


Figure 3. Evolution of the loss function over epochs

The performance of the tested models, subsequent to loading the corresponding weights before overfitting takes place, is summarized in Table 2 through the calculated Precision, Recall and F1-score. In this application the Recall is considered more relevant since disregarding a false negative has a greater impact than responding to a false positive.

Table 2. Bechmarking of the tested models

Model	Precision			Recall			F1-score		
	train	valid	test	train	valid	test	train	valid	test
1	0.70	0.66	0.66	0.69	0.65	0.66	0.68	0.63	0.64
2	0.74	0.68	0.70	0.72	0.67	0.70	0.72	0.66	0.69
3	0.92	0.88	0.87	0.93	0.88	0.86	0.92	0.88	0.86
4	0.84	0.75	0.79	0.83	0.74	0.79	0.83	0.74	0.78

Evaluation

The obtained confusion matrix provides a comprehensive overview of the predictions made by model M3 on the test dataset split compared to the actual ground truth labels, as it can be seen in Figure 4.



Figure 4. Confusion matrix of the best model (M3)

To obtain a comprehensive view of mispredicted images, across the classes with poor performance, we plotted randomly selected images, regarding the TN, FN, FP, TP comparison for the classes in Figure 5.

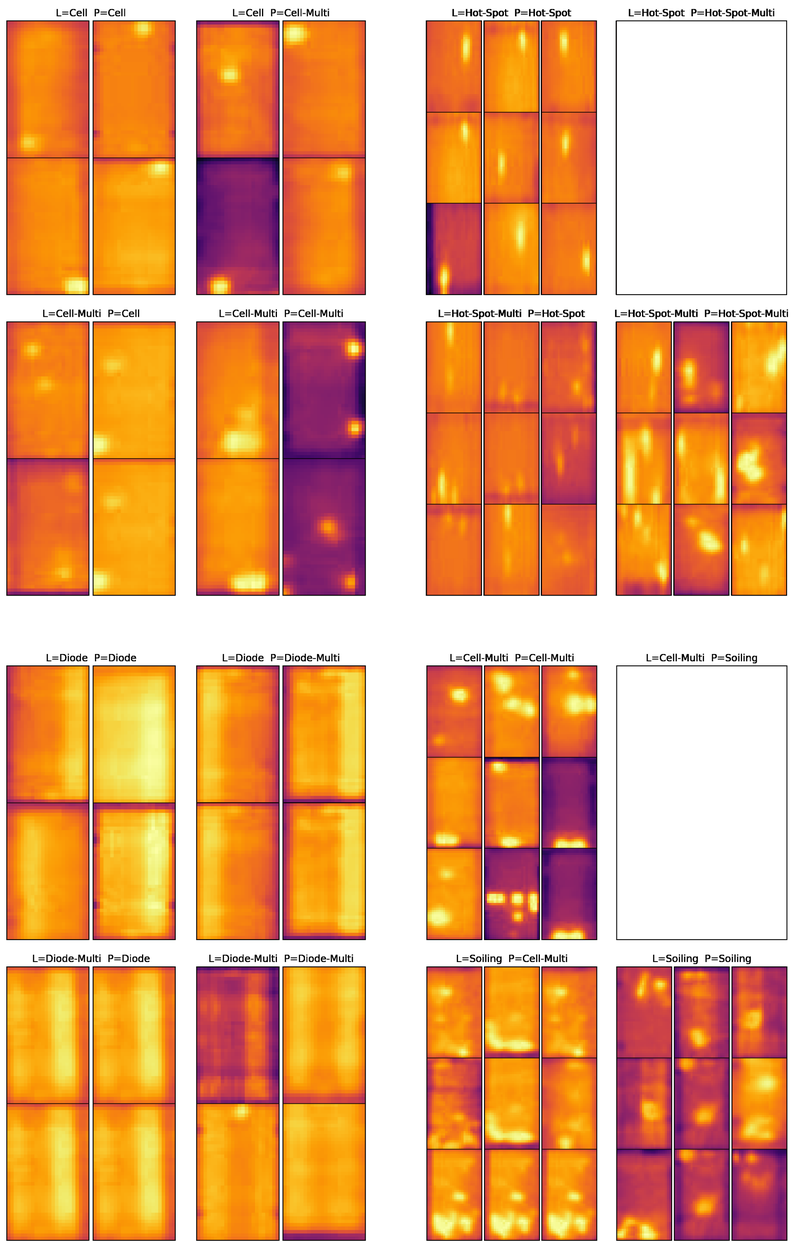


Figure 5. Image comparison of Label against Prediction over different classes

By examining those images, we observe that the models fail to make accurate predictions when the features are not prominent or when have similar geometric patterns.

Conclusions and future work

In this work we investigated the use of simple deep learnig models to classify defects in PV modules through the analysis of UAV collected termographic images. The obtained results demonstrate that the proposed model structures are of high practical value in the context of automated electrical grid asset inspection. In future work we intend to improve the model by test its adaptability with different datasets. More roboust results can be obtained merging thermography analysis with other sensors such as RGB images and V-I signals.

References

[1] Matthew Millendorf, Edward Obropta, and Nikhil Vadhavkar. Infrared solar module dataset for anomaly detection, 2020.

[2] Yahya Zefri, Imane Sebari, Hicham Hajji, and Ghassane Aniba. Developing a deep learning-based layer-3 solution for thermal infrared large-scale photovoltaic module inspection from orthorectified big UAV imagery data. *International Journal of Applied Earth Observation and Geoinformation*, 106:102652, February 2022.